

DEFENSE &
DISASTER
ROBOTICS

SWARM STATE:
5/5 ACTIVE

SECTOR: KEDARNATH VALLEY
(30.73° N, 79.06° E)

AUTONOMOUS MULTI-AGENT SWARM ARCHITECTURE

PROJECT SUTRA

Swarm Unified Tactical Reconnaissance Architecture

A decentralized, fault-tolerant multi-UAV swarm engineered from first principles for **GPS-denied deep mountain valleys**, **-5 dB RF jamming resilience**, and sub-meter survivor geolocation under dense forest canopies.

01. AUTOPILOT & GNC

PX4 50Hz
Offboard,
3D Voxel
Cost-
Volume &
ORCA 3D

02. NEURAL COMMS

Deep JSCC
Autoencoder
with 96.9%
compression
& -5 dB
jamming
survival.

03. AI PERCEPTION

Tri-Modal
RGB/FLIR/mmWave
with 3.59cm
terrain-
corrected
DEM
raycasting.

04. TACTICAL GCS

Pegasus 3D
GIS with 60
FPS
WebGPU
pipeline &
MIL-STD-

Collision
Shield.

2525 ATAK
CoT XML.